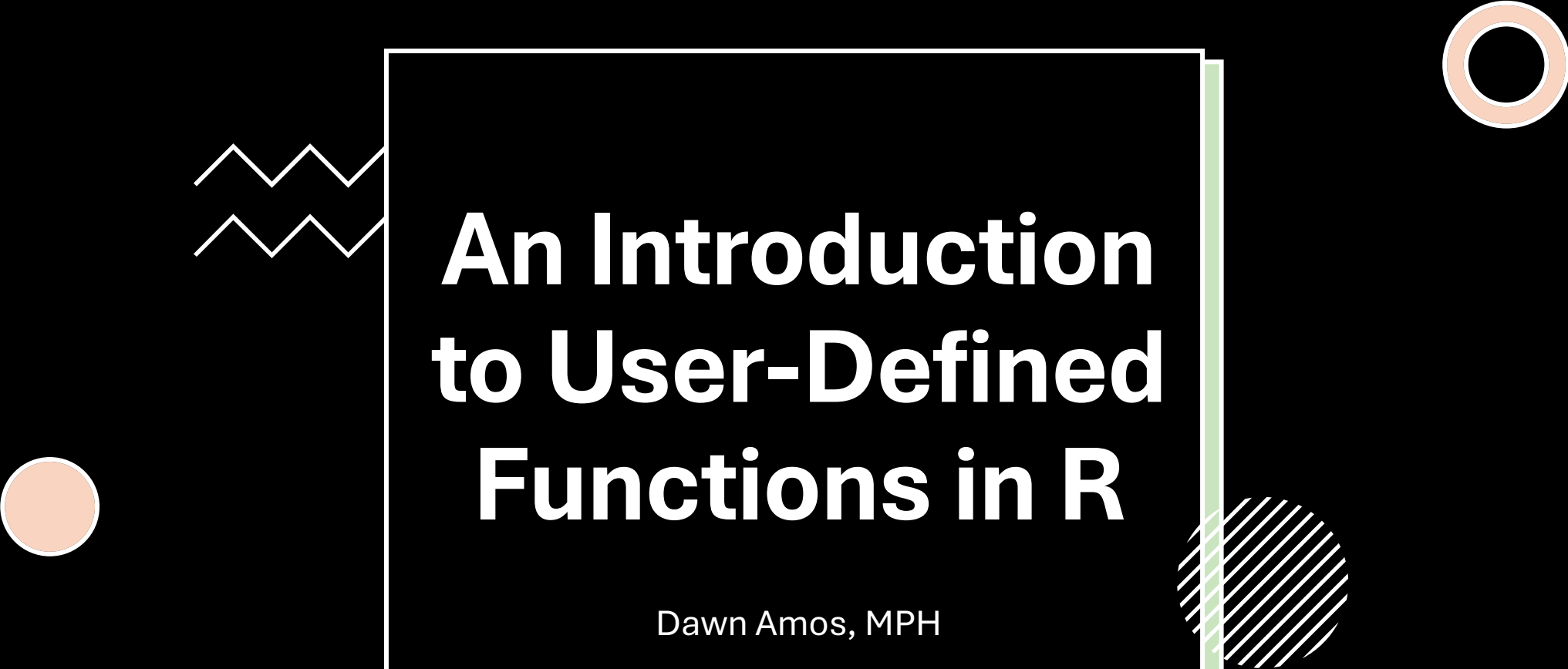


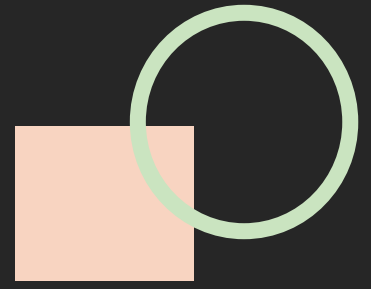


# **An Introduction to User-Defined Functions in R**

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# What are functions in R?



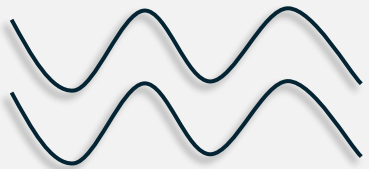
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## 1. Built-in functions: pre-defined

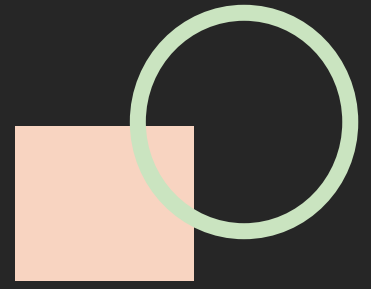
`sum()`

`mean()`

## 2. User-defined functions: created by the user



# What are user-defined functions?

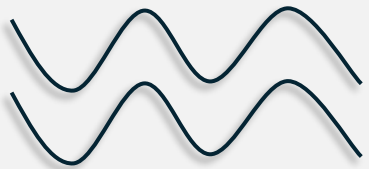


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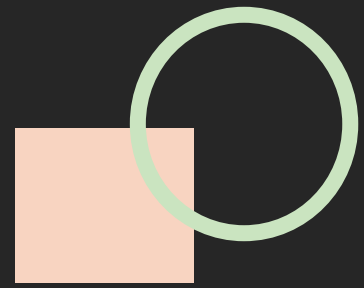
Custom reusable block of code that automates repetitive processes by performing a specific task with provided inputs.

Creating something that tells R:

“Every time I say **x**, do **y** with the arguments I provide.”



# Basic Anatomy of a Function



## Input

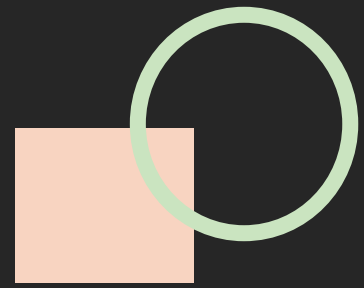
```
# Anatomy of a function
function_name = function(argument_1, argument_2) {
  # code to be executed
  sum(argument_1, argument_2)
}

function_name(10, 10)
```

## Output

```
> # Anatomy of a function
> function_name = function(argument_1, argument_2) {
+   # code to be executed
+   sum(argument_1, argument_2)
+ }
> function_name(10, 10)
[1] 20
```

# Entering Arguments



A function doesn't have to have set arguments.

By not establishing set arguments, it allows us more flexibility with number of inputs.

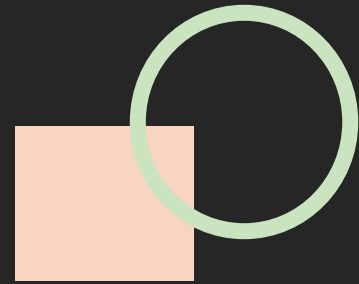
## Input

```
add_function = function(...) {  
  #code to be executed  
  sum(...)  
}  
  
add_function(20, 20, 20, 20)
```

## Output

```
> add_function = function(...) {  
+   #code to be executed  
+   sum(...)  
+ }  
>  
> add_function(20, 20, 20, 20)  
[1] 80
```

# Entering Arguments, cont.



When providing arguments, you can set an argument as a constant.

You can also override that fixed argument.

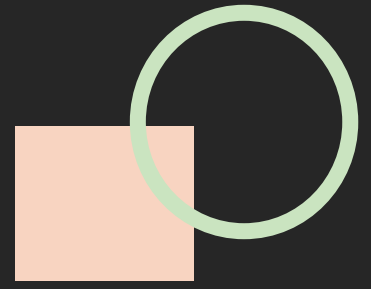
## Input

```
add_20_function = function(x, y = 20) {  
  # code to be executed  
  sum(x, y)  
}  
  
add_20_function(20)  
# override fixed argument  
add_20_function(10, 10)
```

## Output

```
> add_20_function = function(x, y = 20) {  
+   # code to be executed  
+   sum(x, y)  
+ }  
>  
> add_20_function(20)  
[1] 40  
> # override fixed argument  
> add_20_function(10, 10)  
[1] 20
```

# Why use user-defined functions?



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## 1. Saving yourself time

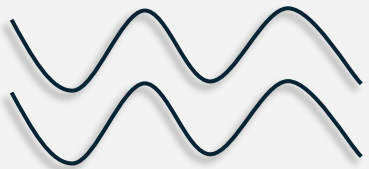
You don't have to rewrite or copy-and-paste blocks of repetitious code.

## 2. Prevents errors

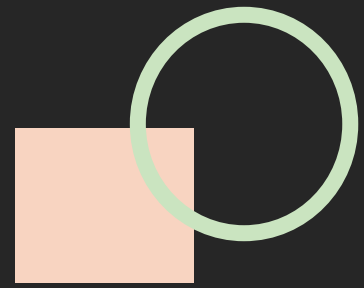
Rewriting and copy-and-pasting blocks of similar code can result in errors with inconsistency, especially if you're still tweaking your code.

## 3. Improve code readability

It makes your code visually cleaner and easier to understand.



# Practice Datasets



| Variable Name | Description                                                       |
|---------------|-------------------------------------------------------------------|
| id            | Unique ID for each sampled mother                                 |
| county_type   | County designation status (rural, remote, urban)                  |
| race          | Race/ethnicity (White, Black, Hispanic, or Other)                 |
| height_in     | Height in inches                                                  |
| hypertension  | Gestational hypertension diagnosis (Yes/No)                       |
| diabetes      | Gestational diabetes diagnosis (Yes/No)                           |
| wt_tri1       | Weight at the end of the first trimester, in pounds               |
| wt_tri2       | Weight at the end of the second trimester, in pounds              |
| wt_birth      | Weight at time of birth, in pounds                                |
| survey_weight | Survey weight, stratified by county and weighed by race/ethnicity |